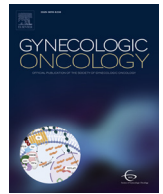




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Review Article

Endometrial cancer: A society of gynecologic oncology evidence-based review and recommendations, part II

Chad A. Hamilton^{a,*}, Bhavana Pothuri^b, Rebecca C. Arend^c, Floor J. Backes^d, Paola A. Gehrig^e, Pamela T. Soliman^f, J. Spencer Thompson^g, Renata R. Urban^h, William M. Burkeⁱ^a Gynecologic Oncology Section, Women's Services and the Ochsner Cancer Institute, Ochsner Health, New Orleans, LA, United States of America^b Department of Obstetrics and Gynecology, NYU School of Medicine, New York City, NY, United States of America^c Division of Gynecologic Oncology, Department of Obstetrics and Gynecology, University of Alabama at Birmingham, Birmingham, AL, United States of America^d Division of Gynecologic Oncology, Ohio State University and James Cancer Hospital, Columbus, OH, United States of America^e Professor and Chief, Division of Gynecologic Oncology, University of North Carolina, Chapel Hill, NC, United States of America^f Department of Gynecologic Oncology and Reproductive Medicine, University of Texas MD Anderson Cancer Center, Houston, TX, United States of America^g Division of Radiation Oncology, Stephenson Cancer Center, University of Oklahoma, Oklahoma City, OK, United States of America^h Department of Obstetrics and Gynecology, Division of Gynecologic Oncology, University of Washington School of Medicine, Seattle, WA, United States of Americaⁱ Department of Obstetrics, Gynecology and Reproductive Medicine, Stony Brook University Cancer Center, Stony Brook, NY, United States of America

HIGHLIGHTS

- Context of recurrence drives treatment options to include combination approaches.
- Surveillance and survivorship should be tailored for endometrial cancer patients.
- Fertility and ovarian preservation can be considered for select patients.
- Primary radiation is reasonable for patients who are not surgical candidates.

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ABSTRACT

In 2014, the Society of Gynecologic Oncology's Clinical Practice Committee published a clinical update reviewing the treatment of women with endometrial cancer. At that time, there had been significant advances in the diagnosis, work-up, surgical management, and available treatment options allowing for more optimal care of affected women.

This manuscript, Part II in a two-part series, includes specific recommendations on treatment of recurrent disease, post treatment surveillance and survivorship, considerations for younger women, and special situations. Part I covered histopathology and molecular pathology, risk factors, presentation and diagnostic approach, surgical approach and adjuvant therapy.

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* Corresponding author.

E-mail address: chad.hamilton@ochsner.org (C.A. Hamilton).

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1. Treatment of recurrent disease

1.1. Clinical question 1

What are the treatment options for pelvic recurrence?

Recommendation 1.1

In patients not previously radiated, pelvic radiation is recommended and can be used for nodal and/or vaginal recurrences. In general, this should be administered as external beam radiation therapy (EBRT), either 3-D conformal or intensity modulated radiotherapy (IMRT), followed by brachytherapy performed using either an intracavitary or an interstitial technique.

Recommendation 1.2

Prior to radiation, surgical resection may be considered for isolated pelvic tumors that are resectable and not considered curable without resection (CII).

Recommendation 1.3

Chemotherapy may be considered for those with high risk of extra-pelvic relapse (CII).

Literature review

Ten to 15% of patients with intermediate risk endometrial cancer experience a pelvic recurrence after surgery alone [1,2]. Adjuvant radiation decreases the risk of pelvic relapse to less than 5% [1–3]. The majority of recurrences (63–75%) will be vaginal [1]. In PORTEC 1, 89% of patients with vaginal relapse who were treated with curative intent, usually EBRT and vaginal brachytherapy (VBT), and occasionally surgery, achieved complete remission. Five-year overall survival after vaginal recurrence in patients without prior radiation was 65%, compared to 43% for patients with a vaginal recurrence who received prior adjuvant radiation. Treating with EBRT prior to brachytherapy treats lymph nodes and paravaginal tissues that may harbor microscopic disease while at the same time shrinking gross vaginal disease allowing lower doses of radiation to surrounding organs. Additionally, image-guided brachytherapy has demonstrated encouraging results potentially through improved target delineation, local control and normal tissue sparing [4]. EBRT or IMRT can also be used effectively to treat nodal recurrences of endometrial cancer and long-term survival can be achieved [5]. In contrast to isolated vaginal recurrence, only 20% of patients with a pelvic relapse treated with curative intent reached complete remission and survival after pelvic relapse is similar to distant recurrence (3-year OS 8% and 14%, respectively) [6].

Treatment of recurrent endometrial cancer after prior radiation is clearly more challenging. With modern imaging and radiation techniques the dose to the target can be optimized while limiting the dose to surrounding critical organs to minimize morbidity and radiation related complications. Several studies have reported on re-irradiation. Ling and colleagues described 22 patients with a vaginal recurrence who previously received vaginal brachytherapy (55%), pelvic radiation (23%) or both modalities (22%). Treatment for vaginal recurrence was with curative intent. With a cumulative rectosigmoid and bladder dose limited to <75 Gy and <90Gy, treatment was overall well tolerated

without grade≥3 radiation related toxicities. Three-year local control was 65% with 41% disease free survival. Most studies report that local control and survival are improved with combined brachytherapy and external beam RT compared with either modality alone [7]. Improved outcomes have also been reported with the use of higher doses of radiation (>80 Gy) [7,8] and image-guided brachytherapy [9–11]. Whether concurrent chemotherapy can further improve disease control and survival is currently being studied in GOG 238 which is a randomized trial of pelvic radiation with or without concurrent weekly cisplatin in women with pelvic-only recurrences of endometrial carcinoma.

While it is encouraging that many vaginal (and some pelvic) recurrences can be salvaged, it is important to note that close surveillance in PORTEC-1 allowed many recurrences to be detected early while still small tumors. Outcomes vary greatly in retrospective studies with and without prior radiation. Two to five-year local control is 44–100%, relapse free survival 26–96%, and overall survival 35–80% [4].

For those patients who have received prior radiation and for whom re-irradiation is not an option, pelvic exenteration may be the only alternative for treatment with curative intent. Unfortunately, 5-year overall survival is 20–60% and the morbidity of the procedure is high [12–15], with the worst outcomes for those with node positive disease.

1.2. Clinical question 2

What are the treatment options for extra-pelvic recurrence?

Recommendation 2.1

Surgery may be considered in select cases where complete surgical resection is feasible and safe. (CII).

Recommendation 2.2

Systemic therapy with either chemotherapy, hormonal therapy, targeted therapy, immunotherapy, or a combination regimen, is recommended for patients with an extra-pelvic recurrence (AI, AII).

Literature review

The role of secondary cytoreductive surgery in recurrent endometrial cancer is not well defined and is mostly based on small retrospective studies and case reports. Bristow et al. identified 35 patients who underwent secondary cytoreductive surgery. Complete cytoreduction was achieved in 66% of patients. Those with no gross residual disease had improved survival (39 months) compared to those with residual disease (13.5 months) which was similar to those treated without surgery (13 months) [16]. Patients with isolated distant metastases such as isolated para-aortic lymph node recurrence may achieve prolonged disease-free survival with surgical resection [17]. Unfortunately, most patients with a distant recurrence present with multifocal disease. While in some cases surgery may be appropriate, most patients will require systemic therapy. Until recently, there were only two Food and Drug Administration (FDA) approved treatment options. These included megestrol acetate hormonal therapy, and pembrolizumab immunotherapy for mismatch repair deficient tumors. Twenty to 30% of endometrial cancers have defective DNA mismatch repair. In 2017, The FDA granted accelerated approval to pembrolizumab (anti PD-1)

as the first tissue agnostic therapy for mismatch repair deficient solid tumors that progressed following prior treatment. A preliminary study including 15 MMR deficient endometrial cancers demonstrated a 53% objective response rate and 73% disease control rate [18]. A recent update showed an ORR 57% and median PFS of 25.7 months amongst 49 patients with dMMR recurrent endometrial cancer [19]. In the KEYNOTE-028 phase IB trial, 24 patients with pretreated PD-L1 positive endometrial cancers, there was a 13% partial response rate with median duration of response not reached and 13% stable disease rate with median duration of response 25 weeks [20].

In September 2019, the FDA approved combination pembrolizumab and lenvatinib for the treatment of recurrent endometrial cancer without microsatellite instability-high (MSI-H) or mismatch repair deficiency (dMMR), who have disease progression following prior systemic therapy and are not candidates for curative surgery or radiation [21].

This was based on the results of the advanced endometrial cancer cohort of the single arm phase Ib/II Keynote-146/Study 111. With extended follow-up, the combination demonstrated favorable efficacy with an overall response rate of 38% and median PFS of 7.4 months. The overall response rate in microsatellite stable disease was 37.2% and 63.6% in MSI-H tumors. Sixty-seven percent of patients experienced grade 3 or 4 treatment-related adverse events and 18% discontinued one or both study drugs due to adverse events [20].

Hormonal therapy is an option to control recurrent disease, especially for patients with low grade endometrial cancer and/or for patients unable to tolerate chemotherapy, however duration of response may be limited. Progestin therapy with megestrol acetate was approved over 40 years ago and is overall very well tolerated. GOG 153 evaluated tamoxifen 20 mg twice daily for three weeks alternating with megestrol acetate 80 mg twice daily for three weeks and reported a 27% response rate. Higher response rates are seen in patients with grade 1 disease compared to grade 3 (38% vs 22%), women less than age 60, and in patients with extra-pelvic disease [22]. Although the median progression free survival was 2.7 months, in 53% of responders the response duration was greater than 20 months. A dosing schedule of tamoxifen 40 mg daily with alternating weekly cycles of medroxyprogesterone acetate 200 mg daily was also beneficial with a 33% response rate [23]. Aromatase inhibitors have minimal activity as single agents [24,25], but in combination, letrozole and everolimus demonstrated encouraging activity with an objective response rate of 24–32% and median duration of response of 15 months [26,27]. Combination letrozole and palbociclib is currently being investigated in the PALEO study through the NSGO, and letrozole and ribociclib through NRG Oncology.

Patients with recurrent endometrial cancer who have not received prior chemotherapy are typically treated with chemotherapy or hormonal therapy. Based on GOG 209, carboplatin and paclitaxel is the standard of care for these patients [28]. For patients with recurrence and prior chemotherapy with a long platinum free interval, combination platinum retreatment is reasonable. Several single agents have been tested in the second line. The GOG-129 single arm phase II series evaluated cytotoxic agents in patients with one prior cytotoxic regimen. Response rates were generally less than 15% except for paclitaxel, which was used prior to adoption of paclitaxel in the front-line setting. The GOG-229 series explored a series of the targeted agents in recurrent and persistent endometrial cancer with modest response rates ranging from 0%–24.5% potentially warranting further investigation [29]. In the MITO END-2 trial, all patients had one prior line of platinum-based chemotherapy and progressed greater than 6 months after completion of prior platinum. This trial showed improved response rate with the addition of bevacizumab to carboplatin/paclitaxel (54% vs 73%) and improved PFS (8.7 vs 13 months, HR 0.57 [0.34, 0.96], $p = 0.036$) [30]. Historically, doxorubicin demonstrated efficacy in endometrial cancer [28]. Although response rates were 19–37% in the first line, when tested in the second line in a phase III trial, doxorubicin demonstrated an

objective response rate of 14%, PFS 4.7 months, and OS 10.8 months [28]. Similarly, liposomal doxorubicin demonstrated limited activity with an overall response rate of 9.5% in previously treated metastatic endometrial cancer in the GOG's phase II trial [31].

1.3. Clinical question 3

What pathologic / molecular testing should be considered to guide treatment?

Recommendation 3.1

Pathologic and molecular testing can help guide treatment decisions and identify (on label) targeted treatment options. Mismatch repair status and/or microsatellite instability testing should be performed on all endometrial tumors to 1) screen for Lynch syndrome, and 2) determine eligibility for (future) single agent immunotherapy use (AI). HER2Neu testing is recommended for serous uterine cancers to determine eligibility for trastuzumab (AI, preferred). Hormone receptor status (estrogen and progesterone receptor status) should be performed to evaluate possible candidates for hormonal therapy (AII). Next generation sequencing may help identify other possible targetable mutations (BII).

Literature review

Hormonal treatment for endometrial cancer makes it one of the first cancers to be treated based on molecular features. As discussed in greater detail in the section on extra-pelvic recurrence, with modern advances, tailored treatment continues to hold great promise. Tumors should be tested for mismatch repair proteins / microsatellite instability based on efficacy of anti PD-1 therapy and an FDA approved therapeutic [18,19]. HER2 immunohistochemistry (IHC) testing (with reflex to HER2 FISH testing for equivocal IHC) should be considered to guide the treatment of advanced stage or recurrent serous endometrial cancer based on compelling phase II data [32]. Next Generation Sequencing is CMS approved for advanced solid tumors and may provide a tool to identify targeted mutations which can help guide treatment and highlight patients who may be eligible for targeted therapy basket trials. Part I of this review further discusses the basis of current molecular pathologic recommendations.

2. Post treatment surveillance and survivorship

2.1. Clinical question 1

What is the appropriate follow-up for women after treatment of endometrial cancer?

Recommendation 1.1

A speculum and pelvic examination, in addition to a review of systems to elicit any new symptoms associated with a possible recurrence, should be completed every 3–6 months for 2 years, and every 6–12 months thereafter in patients with endometrial cancer (CIII). It is acceptable to follow patients with low-risk endometrial cancer with less frequency (e.g. every 6–12 months for first 2 years, then yearly thereafter) (CIII).

Recommendation 1.2

Do not perform cytology (Pap tests) of the vaginal cuff in patients with a history of endometrial cancer and no prior history of high-grade cervical dysplasia (CII).

Recommendation 1.3

Imaging (e.g. CT scans, PET/CT scans) should be used if there is a suspicion for recurrent disease (CIII).

Literature review

In addition to detecting treatable, recurrent disease, surveillance provides psychosocial reassurance and may improve quality of life. Prospective data does not guide current recommendations. Given that most

endometrial cancers are early stage when initially diagnosed and treated, and that recurrence is often local and curable, a cost-effective surveillance strategy is desirable. An institutional review of the incorporation of the initial 2011 SGO surveillance guidelines, which focused on physical exam and symptoms as primary surveillance, reported equivalent outcomes with appreciable decrease in surveillance costs [33]. Separate surveillance guidelines set forth by the NCCN and SGO are largely congruent, and the SGO guidelines offer a stratified approach based on low and high-risk characteristics [34]. Although the current NCCN guidelines recommend a physical examination every 3–6 months for 2–3 years, then 6 months or annually it is reasonable for patients with low-risk endometrial cancer to be followed with less frequency [34]. The SGO review recommends a thorough speculum and pelvic examination in addition to a review of systems to elicit any new symptoms associated with recurrence, such as vaginal bleeding, abdominal or pelvic pain, weight loss, headaches, coughing, or lethargy [34]. NCCN supplements this list with bladder or rectal bleeding, decreased appetite, shortness of breath, and swelling in the abdomen or legs.

The SGO recommends against vaginal cytology to aid in the detection of recurrence at the vaginal cuff. Most vaginal recurrences are detected with clinical examination alone, and asymptomatic recurrences are infrequently detected with vaginal cytology [35–37]. The NCCN and SGO recommend that radiologic evaluation such as CT or PET/CT scans be used only if there is concern for recurrence. However, the NCCN notes that for patients with treated stage III–IV disease, CT of the chest, abdomen, and pelvis every 6 months is an option [36]. The NCCN and SGO recommend that CA-125 may be used in surveillance for those patients who have an elevated CA-125 prior to treatment. The SGO notes that the use of CA-125 may also be appropriate in patients with advanced disease or serous endometrial cancer [34].

Although patients may prefer that surveillance care be provided by an oncologist, it is also safe and reasonable for patients with low-risk endometrial cancer to be followed by a gynecologist once two years have elapsed from their treatment [36]. Patients with advanced stage disease and/or high-risk histologic types should be followed by a gynecologic oncologist until five years have elapsed since their treatment, although alternating visits can be considered.

2.2. Clinical question 2

Are there survivorship issues unique to endometrial cancer patients?

Recommendation 2.1

Following treatment, endometrial cancer patients should be counseled on the impact of obesity, lifestyle and nutrition (CIII).

Literature review

One of the most common risk factors for endometrial cancer is obesity, therefore obesity-related comorbidities should be addressed in the survivorship period. Highlighting this, cardiovascular disease is the leading cause of death in endometrial cancer survivors making interventions to address cardiac risk factors a necessary part of a survivorship care plan [38]. Although weight and obesity have not been found to impact the risk of endometrial cancer recurrence [39], obesity affects both quality of life and overall survival.

3. Endometrial cancer considerations for younger women

3.1. Clinical question 1

How should patients considering fertility-sparing options be evaluated?

Recommendation 1.1

Patients who desire fertility sparing treatment should be evaluated by D&C (Preferred) or endometrial biopsy to evaluate grade. (All).

Recommendation 1.2

An MRI to evaluate for myometrial invasion, lymphadenopathy, and adnexal pathology should also be completed (All).

Literature review

Although the majority of endometrial cancer patients are postmenopausal, up to 14% of women are premenopausal, and 4% younger than 40 years old [40,41]. This trend is expected to increase with the increasing prevalence of obesity, and though younger women are more likely to be stage I and have low-grade disease, 20% will still be diagnosed with disease beyond the uterus and over half with higher grade disease [41]. This poses a unique treatment challenge in this population where fertility preservation, ovarian conservation, and long-term treatment toxicity must be balanced with the oncologic risk of non-standard treatment [42,43].

The most common risk factors for the development of endometrial cancer in young women are increasing body mass index (BMI), nulliparity, and irregular menstrual cycles [44]. Polycystic ovary syndrome, a common cause of anovulatory cycles associated with excessive endogenous estrogen, has also been associated with an increased risk of endometrial cancer [45,46]. The risk for developing endometrial cancer may be increased as much as 22-fold in women younger than 45 years of age with BMIs that are greater than 35. There may be an increased rate of mutations associated with Lynch syndrome in young women with endometrial cancer. A prospective multi-institutional study found a 9% rate of germline mutations associated with Lynch syndrome in women younger than 50 years old who developed endometrial cancer [47]. Predictors of germline mutation were a first-degree relative with a Lynch-syndrome associated cancer, tumor with loss of MSH2 expression, tumors with high microsatellite instability and lower BMI. Currently, the SGO and American College of Obstetricians and Gynecologists recommend a systematic approach to identifying women with Lynch syndrome that includes either selective or universal tumor testing of endometrial carcinomas for MMR proteins or MSI testing [48].

While hysterectomy is considered standard of care for endometrial cancer, fertility-sparing options should be considered with appropriate counseling for a young patient desiring future fertility. Data on long-term and pregnancy-related outcomes are limited by small sample size. A key principle of the evaluation of patients is ensuring the presence of a grade 1 endometrioid tumor limited to the endometrium; such tumors have a lower risk of extrauterine or nodal metastases. In a Swiss registry study, only 18% of endometrial cancer patients aged 45 years or younger had stage IA, grade 1 disease [49]. For women who wish to pursue fertility-sparing options, dilation and curettage (D&C) is preferred for evaluating the tumor grade. In a review of more than 1400 cases of endometrial cancer, post-hysterectomy grade was higher in 8.7% of cases diagnosed by D&C which was significantly lower compared to 17.4% diagnosed by office endometrial sampling [50]. MRI is preferred for assessment for myometrial invasion and adnexal pathology or alternatively, transvaginal ultrasound if MRI is not available. A meta-analysis demonstrated that MRI has a better sensitivity than ultrasound for detecting deep myometrial invasion [51]. Several authors and the SGO recommend assessment of tumor progesterone status to better identify ideal candidates for conservative treatment [52]. Though there is limited data from fertility-sparing series and reports, a historical review reported response rates to progestin therapy of 72% for progesterone-positive tumors and 12% for progesterone-negative tumors [53]. In parallel with the above work-up, the clinician and patient, in the process of shared decision making, should discuss the likelihood of attaining pregnancy, potentially with consultation from an infertility specialist.

3.2. Clinical question 2

What is the recommended fertility-sparing treatment of endometrial cancer?

Recommendation 2.1

Progestin therapy with oral or intrauterine progestins should be used (AII).

Literature review

Progestins have been the mainstay of conservative hormonal treatment for endometrial cancer in young woman who want to preserve fertility. These progestin-based therapies include the levonorgestrel-releasing intrauterine system (L-IUS), medroxyprogesterone acetate (MPA) and megestrol acetate. Others have suggested combined oral and intrauterine progestins [54] or hysteroscopic resection followed by L-IUS placement [55]. In a systematic review of oncologic outcomes in patients with grade 1 adenocarcinoma treated with progestins, the complete response rate was 48.2%. The risk of recurrence after initial response was 35.4% in women with carcinoma; persistent disease was noted in 25.4% [56]. In patients with hyperplasia, a review and meta-analysis noted that the response rate with the levonorgestrel IUS was higher for complex and atypical hyperplasia compared with oral progestins [57]; however, in a review of patients with complex atypical hyperplasia or early endometrial adenocarcinoma, the treatment with oral or intrauterine progestin was noted to be similarly effective. [58]

3.3. Clinical question 3

How long can a patient be treated conservatively before treatment is considered unsuccessful?

Recommendation: 3.1

Fertility sparing treatment for endometrial cancer is typically continued for 6–12 months (BII).

Recommendation 3.2

Patients will need to be thoroughly counseled about the risks and benefits of fertility sparing endometrial cancer treatment (BIII).

Literature review

The optimal regimen, route and follow-up of progestin therapy is not well defined, as most available data are limited to retrospective series and reviews. Following the initiation of progestin-based therapy, it is recommended to repeat sampling with office biopsy or D&C in 3–6 months. In the review by Gunderson et al., the median time to complete response to progestin therapy was 6 months [56]. Hysterectomy with staging should be considered once childbearing is complete, if patients have documented progression on biopsies, and/or of endometrial cancer is still present after a specified duration of progestin-based therapy. A recent review of patients <45 years of age from the SEER database revealed no difference in survival for patients treated with progestin compared to those undergoing hysterectomy [59].

A systematic review of reproductive outcomes for patients with grade 1 adenocarcinoma treated with progestins showed that nearly 35% of those with a history of carcinoma became pregnant [56]. A case series accompanied by a systematic review noted 65 deliveries with 77 live births; these pregnancies resulted from both assisted reproductive technologies and spontaneous conceptions. One maternal death was seen due to recurrent disease [60]. Though pregnancy outcomes have been promising, it is important each patient understand the high likelihood of needing reproductive technology. Consultation with a Reproductive Endocrinologist concomitant with beginning medical therapy is advised [61].

3.4. Clinical question 4

Can ovarian preservation be considered?

Recommendation 4.1

Ovarian preservation may be considered in premenopausal women with low grade, early stage endometrial cancer with normal appearing adnexa and no evidence of extra-uterine disease (CII).

Recommendation 4.2

Risk-reducing hysterectomy and bilateral salpingo-oophorectomy is the recommended risk-reducing option for women with Lynch syndrome who have completed childbearing with a suggested age of 40 to 45 years old. For women considering hysterectomy with ovarian preservation, complete resection of the fallopian tubes is recommended. (AII).

Literature review

Traditionally, surgeons perform bilateral salpingo-oophorectomy (BSO) in conjunction with hysterectomy to detect occult disease and reduce recurrence risk from continued estrogen production. With an evolving knowledge of the health risks of oophorectomy in premenopausal women [62], there is interest in the impact of ovarian preservation. A recent institutional review showed that in patients with pelvic-confined disease, only 0.8% had microscopic ovarian involvement; all patients with ovarian involvement had FIGO grade 3 disease, deep myometrial invasion and extrauterine involvement of either cervix or lymph nodes [63]. The Korean Gynecologic Oncology Group demonstrated that in a group of women with stage I-II endometrial cancer who had ovarian preservation, there were no recurrences in patients with stage IA disease [64], and ovarian preservation did not have an impact on either recurrence-free or overall survival [65]. Similarly, an analysis of ovarian preservation at the time of hysterectomy for women with early stage endometrial cancer using the National Cancer Database found no excess deaths associated with ovarian preservation [66]. Accumulating data including recent systematic reviews appear to confirm the safety and perhaps benefits of ovarian preservation in younger patients with early-stage endometrial cancer [56]. Earlier reports raised concern for synchronous ovarian malignancy which was reported in a younger patient population to be as high as 25% [67,68]. However, the majority of these patients either had identifiable extrauterine disease at the time of surgery or gross abnormalities of the ovary. For women without gross extrauterine disease and normal appearing adnexa, the risk of ovarian involvement appears to be less than 1% [69].

The recommendation for risk-reducing hysterectomy combined with bilateral salpingo-oophorectomy in young women with Lynch Syndrome largely stems from a study of a retrospective cohort of 315 women with Lynch syndrome. In this study 61 and 47 women underwent risk-reducing hysterectomy and risk-reducing bilateral salpingo-oophorectomy respectively and they were matched with mutation positive women who had not undergone risk reducing surgery. The incidence of both endometrial and ovarian cancer was significantly lower in the patients who had risk reducing surgery. No patients in the surgery groups developed endometrial or ovarian cancer compared to 33% and 5% in the non-surgery group who developed endometrial and ovarian cancer respectively [70]. Risks and benefits of ovarian preservation should be thoroughly discussed inclusive of the implications of surgically induced menopause and the limited data at present regarding lifetime risk of ovarian cancer with specific mutations. If the ovaries are preserved, the complete fallopian tubes should be removed as fallopian tube cancers have been reported [48,71].

3.5. Clinical question 5

Are patients with a personal history of endometrial cancer candidates for estrogen therapy for menopausal symptoms?

Recommendation 5.1

In patients with low grade, early stage endometrial cancer, hormonal therapy can be considered for postmenopausal patients with severe menopausal symptoms not otherwise relieved (CII).

Literature review

Historically the use of estrogen therapy after endometrial cancer treatment raised concerns for increasing risk of cancer recurrence. This concern has largely been resolved by studies demonstrating

the safety of estrogen therapy in postmenopausal women with endometrioid endometrial cancer [72]. In GOG 137, patients with stage I-II endometrial cancer were randomly assigned to estrogen therapy. Due to the publication of the Women's Health Initiative during the study period, the study was terminated prior to meeting its accrual goal. The overall recurrence rate was 2% and although the study could not statistically refute or support the use of estrogen, the investigators concluded that in patients with low grade stage I-II endometrial cancer, hormone therapy appears to be associated with a low risk of endometrial cancer recurrence. Additional case-control studies support these findings [73,74]. Although a meta-analysis notes the safety of hormone replacement [75], there are limited data on the use of hormone replacement in patients with Type II and/or advanced stage endometrial cancers.

4. Special situations

4.1. Clinical question 1

Should surgical staging be completed in all patients who have an incidental diagnosis of endometrial cancer following hysterectomy for another indication?

Recommendation 1.1

Women found to have endometrial cancer incidentally after hysterectomy should have their risk for extrauterine disease and potential for disease recurrence evaluated based on age, histologic cell type, and uterine tumor features. Individualized treatment plans can be based on the findings (B).

Literature review

The need for repeat surgery for the sole purpose of staging in women discovered to have endometrial cancer following a hysterectomy should be considered carefully. A dedicated study will probably never be performed because of the relative rarity of the situation. Comprehensive pathology review is mandatory in order to retrieve as much information as possible about the features of the uterine cancer, including histologic cell type, FIGO grade, depth of myometrial invasion, presence of lymphovascular space invasion, and tumor size. If such a patient has a tumor with endometrioid histology, grade 1 or 2 tumors, small tumor volume, and superficial myometrial invasion, further intervention may not be indicated after hysterectomy because these features are compatible with a low risk of extrauterine disease and recurrence [76]. Patients who have intermediate- or high-risk features for extrauterine spread or recurrence, patients with high-risk histologic cell types, and older patients should be considered for comprehensive surgical staging. If the patient is a good candidate for surgery, comprehensive staging can be beneficial either by helping avoid unnecessary adjuvant therapies or by guiding such therapies.

If the patient is not a good surgical candidate and/or has uterine features suggestive of intermediate to high risk for extrauterine disease or disease recurrence, imaging along with CA 125 can be used to evaluate for extrauterine disease. The addition of PET to CT scan has been found to have improved sensitivity for detecting nodal metastases while maintaining the specificity of CT scan [77]. Adjuvant radiation and/or chemotherapy can be administered based on the outcome of the diagnostic evaluation.

4.2. Clinical question 2

Can radiotherapy be used as a primary treatment modality for endometrial cancer?

Recommendation 2.1

Select women diagnosed with endometrial cancer who are not candidates for surgery can be treated with primary radiation therapy. Some patients may also benefit from chemotherapy (BII).

Literature review

In patients who cannot undergo hysterectomy or surgical staging following a diagnosis of early stage endometrial cancer, primary RT with external beam radiation and intracavitary radiation remains a reasonable option for loco-regional disease control. Several studies evaluated patients in this unique circumstance. The five-year overall survival following primary radiation therapy ranges from 39 to 71% [78,79] but locoregional control and cancer-specific survival can be excellent with rates of 90% and 86% respectively in one study of 45 inoperable clinical stage I patients treated with high-dose rate image-guided brachytherapy [80,81].

Imaging may be used to assess for extrauterine disease. This may allow for consideration of palliative chemotherapy following completion of radiation therapy, or for determination of metastatic disease that may benefit from palliative radiation. Those women who have been diagnosed with high-risk histology, such as grade 3 endometrioid, clear cell, serous and carcinosarcoma, should be considered for palliative chemotherapy. In addition, molecular studies may identify patients who may benefit from palliative treatment with hormones or immunotherapy.

Author contributions

Chad Hamilton: methodology, writing-original draft, writing-review and editing, supervision, project administration.

Bhavana Pothuri: writing- review and editing.

Rebecca Arend: conceptualization, writing-original draft, writing-review and editing.

Floor Backes: conceptualization, writing-original draft, writing-review and editing.

Paola Gehrig: conceptualization, writing-original draft, writing-review and editing.

Pamela Soliman: conceptualization, writing-original draft, writing-review and editing.

J. Spencer Thompson: conceptualization, writing-original draft, writing- review and editing.

Renata Urban: conceptualization, writing-original draft, writing- review and editing.

William Burke: Conceptualization, methodology, writing-review and editing, supervision.

Declaration of Competing Interest

The authors of this paper report that they have no conflicts of interest related to any of the content of this manuscript.

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